

GitHub Link: <https://github.com/pmehrotra3/repo759/HW01>

Task 1.

I went through a) through c) and understand how to time code, how to submit my assignments with git, and what the recommended workflow is when it comes to working on my assignment.

Task 2.

- a) `cd somedir`
- b) `cat sometext.txt`
- c) `tail -n 5 sometext.txt`
- d) `tail -n 5 *.txt`
- e) `for i in {0..6}; do echo $i; done`

Task 3.

- a) No modules have been loaded.
- b) gcc (GCC) 14.3.1 20251022 is available to us.
- c) The available cuda modules are:
`nvidia/cuda/10.2.2, nvidia/cuda/11.0.3, nvidia/cuda/11.3.1, nvidia/cuda/11.6.0, nvidia/cuda/11.8.0, nvidia/cuda/12.0.0, nvidia/cuda/12.1.0, nvidia/cuda/12.2.0, nvidia/cuda/12.5.0, nvidia/cuda/12.9.1, nvidia/cuda/13.0.0 (Default)`
- d) LAMMPS is a piece of software that has a module available on the Euler cluster. It is a parallel particle simulator that can simulate anything from single atoms up to bulk materials.
Reference: <https://www.lammps.org/#gsc.tab=0>

Task 4.

`task4.sh` present in HW01 folder.

Task 5.

- a) A Slurm job begins execution in the same directory in which the command was issued.
- b) `SLURM_JOB_ID` is a unique numeric that identifies a particular job submitted to Slurm.

- c) I would be using the command `squeue` to track the status of the jobs run by me.
- d) I would first find the job id of the job I want to cancel by running the command `squeue -u <your_username>`, then I would cancel it using `scancel <job_id>`.
- e) The command specifies that the job requires one GPU unit for its execution.
- f) The command runs the script as a job array of 10 tasks, indexed from 0 to 9 (inclusive). All the tasks run the same script and have the same arrayjob id, but each has its own value of `SLURM_ARRAY_TASK_ID` (0 through 9), which can be used inside the script to make each task perform different work.

Task 6.

`task6.cpp` present in HW01 folder.